

Detecting Fake messages

We address the problems of Fake news/messages using NLP techniques. In this tutorial we classify whether a given message/ text is Real or Fake Message.

The aim is to use a Bag of n-grams to pre-process the text and apply different classification machine learning algorithms.

The dataset we work on is available at :

<https://www.kaggle.com/datasets/clmentbisailon/fake-and-real-news-dataset>

1. Using Pandas, give all the informations on the dataset : number of lines - columns - classes, etc.
2. Checks if the classes are balanced or not. Create a label column.
3. Create a 'train-test' set with test size of 20%
4. print the shapes of the training and testing sets : X_train and X_test
5. Create the following pipeline object : Trigram vectorizer (1,3) -> KNeighbors classifier (10 neighbours) with "euclidian" metric. Hint use Pipeline([...]) function.
6. Using the pipeline, train the dataset by fitting X_train and y_train
7. Print the classification results Comparison between predictions and targets.
8. Same question with the pipeline : Trigram vectorizer (1,3) -> KNeighbors classifier (10 neighbours) with "cosine" metric.
9. Same question as above with the pipeline object : Trigram vectorizer (1,3) -> RandomForest Classifier.
10. Try other pipelines.